

Project Proposal, Variable B: Dynamic Asset Allocation and Robustness to Geopolitical Shocks

A Comparative Analysis of Markowitz, Michaud, and Black–Litterman

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Context and Research Objective

The classic Mean-Variance optimization (Markowitz, 1952) represents the pillar of portfolio theory, but its practical *out-of-sample* implementation is severely compromised by the *estimation error amplification* phenomenon (Michaud, 1989). Due to the inherent uncertainty in expected return estimates, quadratic optimizers tend to generate corner solutions, maximizing statistical noise and causing unacceptable turnover in unstable market regimes.

The objective of this project is to design, implement, and evaluate *Flexible Asset Allocation* algorithms that resolve this fragility. Specifically, we compare the classical Mean-Variance benchmarks with Michaud's *Resampled Efficiency* and the Bayesian framework of the Black–Litterman model, testing their empirical robustness during periods of severe geopolitical and systemic stress.

Original Initiative: Stress-Testing the Middle East Escalation

As an original proposal compared to the baseline framework, this study includes a dedicated *event study* focused on the recent geopolitical escalation in the Middle East: the initial shock of **October 7, 2023**, and the subsequent **Strait of Hormuz crisis (February/March 2026)**. We evaluate how different allocation strategies react to these sudden *regime shifts* in asset correlations. In this context, the Black–Litterman model is deployed to tactically integrate distinct directional *views*, operationalized as a robust overlay starting from the implied equilibrium returns derived via reverse optimization.

1. Data Universe and Rolling Backtest Protocol

The empirical analysis is conducted on a custom multi-asset universe comprising 22 ETFs, selected to ensure comprehensive coverage across global equities, fixed income duration, credit, real assets, and cash proxies.

- **Data Specifications:** Historical end-of-month log-returns are extracted via `yfinance` spanning from January 2010 to May 2026, yielding over 15 years of continuous data.
- **Backtesting Protocol:** We implement a monthly rolling-horizon framework with a fixed 60-month estimation window to compute time-varying joint moments (μ, Σ) . This guarantees that all strategies are executed purely out-of-sample using only information available at each rebalancing date.

- **Operational Constraints:** Optimization routines across all models enforce strict long-only ($0 \leq w_i \leq 1$) and fully invested ($\sum w_i = 1$) budget constraints, ruling out leverage and short-selling to reflect standard institutional mandates.

Table 1. Asset Class Mapping and ETF Universe

Ticker	Asset Class / Description	Ticker	Asset Class / Description
SPY	S&P 500 (US Large Cap)	EMB	Emerging Market Bond (USD)
QQQ	Nasdaq-100 (US Growth)	LQD	Investment Grade Corporate
VGK	MSCI Europe	HYG	High Yield Corporate Bonds
EWJ	MSCI Japan	GLD	Gold Bullion
VPL	MSCI Asia-Pacific	SLV	Silver
EEM	MSCI Emerging Markets	DBC	Broad Commodity Index
VWO	FTSE Emerging Markets	XLE	Energy Select Sector (Equity)
SHY	1–3Y US Treasury (Short Duration)	TIP	US Inflation-Protected (TIPS)
IEF	7–10Y US Treasury (Intermediate)	VNQ	US Real Estate (REIT)
TLT	20+Y US Treasury (Long Duration)	BIL	1–3M T-Bill (Risk-Free Proxy)
IGOV	International Government Bonds	USO	United States Oil Fund (Crude)

2. Implementation of Asset Allocation Models

To isolate the effects of estimation error and parameter optimization, we deploy four distinct allocation routines within the same backtesting engine.

2.1. Classical Mean-Variance: Target Expected Return Portfolio

The first model implements the standard Markowitz tracking optimization by minimizing portfolio variance subject to a fixed target expected return (μ_{target}). This portfolio acts as a baseline benchmark to capture the structural tendency of traditional quadratic optimization to construct extreme, concentrated allocations based entirely on ex-post historical averages.

2.2. Classical Mean-Variance: Maximum Sharpe Ratio Portfolio

The second configuration solves for the tangency portfolio by maximizing the ex-ante Sharpe Ratio, defined as the ratio of excess returns over the risk-free rate proxy ($r_f = 4.0\%$, approximating the Federal Funds rate environment) to portfolio volatility. To visually demonstrate the structural instability of this portfolio to sample variations, a preliminary perturbation analysis ($\pm 1\text{--}2\%$ random shock on μ) is embedded, showing the violent reshuffling of the classical weight frontiers.

2.3. Resampled Efficiency: The Michaud Framework

To mitigate the optimization fragilities of the previous structures, the third model implements Michaud’s Resampled Efficiency. At each rebalancing step, the engine runs Monte Carlo simulations ($M = 100$ independent paths) drawing from the predictive distribution of the parameters over a $K = 30$ point frontier. By averaging the resulting optimal vectors, this statistical regularization eliminates discrete corner solutions, forcing a smooth and highly diversified transition of asset weights over time.

2.4. The Bayesian Tactical Overlay: Black–Litterman Setup

The final model integrates forward-looking subjective beliefs concerning the 2026 Strait of Hormuz crisis into the CAPM market implied prior distribution $\mu \sim \mathcal{N}(\Pi, \tau\Sigma)$. The equilibrium excess return vector is extracted via reverse optimization as $\Pi = \delta\Sigma w_{eq}$, where the aggregate risk aversion coefficient δ is derived endogenously from the market portfolio proxy (w_{eq} based on ETF AUM market proxies).

To evaluate macroeconomic distress under this supply shock, the model formalizes four distinct tactical views within the standard P and Q matrix framework:

1. **Energy Supply Shock Leg:** $USO - SPY = +12.0\%$ annualized, capturing immediate crude supply scarcity alongside cost-push inflationary stress on US equities.
2. **Flight-to-Safety Leg:** $GLD - EEM = +9.0\%$ annualized, positioning gold as a safe haven while emerging markets experience capital outflows.
3. **Sectoral Pass-Through Leg:** $XLE = +12.0\%$ absolute annualized excess return, assuming direct margin pass-through to energy producer earnings.
4. **Asymmetric Macro Leg:** $SPY - VGK = +4.0\%$ annualized, reflecting US domestic shale production resilience versus Eurozone energy import dependency.

The performance differentials (Q) are calibrated conservatively at roughly one-third of the magnitude observed during comparable historical shocks (e.g., 1990 Gulf War, 2003 Iraq War, 2019 Aramco attacks) to prevent destabilizing weight concentrations.

The prior uncertainty scalar is calibrated to a baseline of $\tau = 0.025$, following He and Litterman (1999) and Drobetz (2001). To formalize view uncertainties without injecting arbitrary subjectivity, the diagonal covariance matrix Ω is calibrated using the native variance-proportional approach:

$$\Omega = \text{diag} \left(P (\tau \Sigma) P^\top \right)$$

This specification links the confidence of each view directly to the historical covariance structure of the underlying assets. Finally, to evaluate model robustness, continuous sensitivity sweeps are executed by varying τ ($\in [0.005, 0.10]$) and scaling Ω by a factor c ($\in [0.1, 10.0]$), tracing the smooth boundary transition between the CAPM prior and the active tactical overlay.

3. Out-of-Sample Performance and Operational Evaluation

The realized performance is tracked from January 2015 to May 2026. Crucially, all model returns are penalized by a transaction cost haircut proportional to their monthly portfolio shifts to assess their practical operational feasibility.

Comparative compounding performance is visualized using Cumulative Wealth curves (Equity Line), while downside patterns are examined via Underwater Plots to inspect drawdown depth and recovery duration during major shock periods. Finally, a comprehensive KPI matrix contrasts the realized annualized Sharpe and Sortino ratios, Maximum Drawdown, the Calmar index, and the annualized portfolio Turnover, testing whether robust statistical smoothing and bayesian tactical overlays deliver real alpha against a naive $1/N$ asset benchmark.

Main Reference Bibliography

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